

Term Project Report

CMP2004 Advanced Programming

Project Details	
Project Title	Information for Everyone (Supportive Information App)
Selected Option	Option 3: Supportive Information App
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1. Purpose of the Application

This application provides disadvantaged individuals with a highly accessible, free, and intuitive platform to obtain critical information. It serves as a digital guide, offering curated advice and resources across three vital categories: **Health Information**, **Legal Rights**, and **Social Support**. By prioritizing an accessible interface, the application ensures that users experiencing technical or cognitive barriers can effortlessly access the essential assistance they need.

2. Main Features

- **Category-Based Navigation:** Information is organized into distinct, easily identifiable categories (Health, Legal, Social) using large iconography and highly legible text.
- **Real-Time "Tip of the Day":** The application connects to a live internet API to fetch and display dynamic, real-time advice directly on the category pages.
- **System Notifications:** An automated background service triggers a system notification to remind users of daily tips and available resources.
- **Accessibility & Material Design:** The UI is built on Material Design 3 principles, featuring elevated CardView elements, smooth list animations, and high-contrast color palettes to maximize

readability.

- **Emergency and Feedback Tools:** Includes one-tap access to emergency services (112) and an integrated email intent system for streamlined user feedback.

3. Technical Structure

The application was developed in Android Studio using Java, targeting **Android API Level 36**.

The technical foundation comprises the following core components:

- **Object-Oriented Design:** Employs a structured architecture utilizing Models (`InfoCategory`, `InfoItem`), an Adapter (`InfoItemAdapter`), and a Singleton repository (`InfoRepository`) for robust local data management.
- **Activity & Intent Management:** Contains five distinct activities managed with strict adherence to lifecycle awareness. Navigation is handled via explicit intents paired with custom transition animations.
- **Networking:** Implements asynchronous HTTP requests using `HttpURLConnection`(via `HttpRequestTask`) to fetch JSON data from `api.adviceclip.com/advice` without blocking the main UI thread.
- **Background Services:** Utilizes a standard Android `Service`(`TipNotificationService`) coupled with a `BroadcastReceiver`(`BootReceiver`) to reliably handle system-level notifications.
- **Modern Permissions:** Implements dynamic runtime permissions (`ActivityCompat.requestPermissions`) to properly accommodate the `POST_NOTIFICATIONS` requirement introduced in recent Android versions.

4. Challenges and Solutions

Challenge 1: Main Thread Blocking

Issue: Handling network requests without freezing the user interface or causing Application Not Responding (ANR) crashes.

Solution: The HTTP request logic was decoupled from the main thread using an asynchronous background task. Once the JSON response is successfully received and parsed,

`runOnUiThread`

is safely invoked to update the "Tip of the Day"
TextView
on the active screen.

Challenge 2: System Notification Failures

Issue: System notifications were failing to trigger on newer Android devices (API 33+).

Solution: Because modern Android security policies strictly block notifications by default, a runtime permission check was integrated into the

HomeActivity

's

onCreate

method. The application now explicitly prompts the user to grant the

POST_NOTIFICATIONS

permission upon initial launch, ensuring the notification service functions reliably.

5. Conclusion

The *Information for Everyone* application successfully fulfills all requirements for the CMP2004 term project. It demonstrates the practical implementation of core Android development concepts, including UI design, Activity lifecycles, background services, and internet connectivity. Beyond its technical execution, the project successfully delivers a socially meaningful tool that is polished, accessible, and highly beneficial for its target demographic.